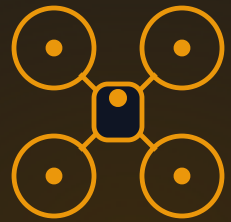
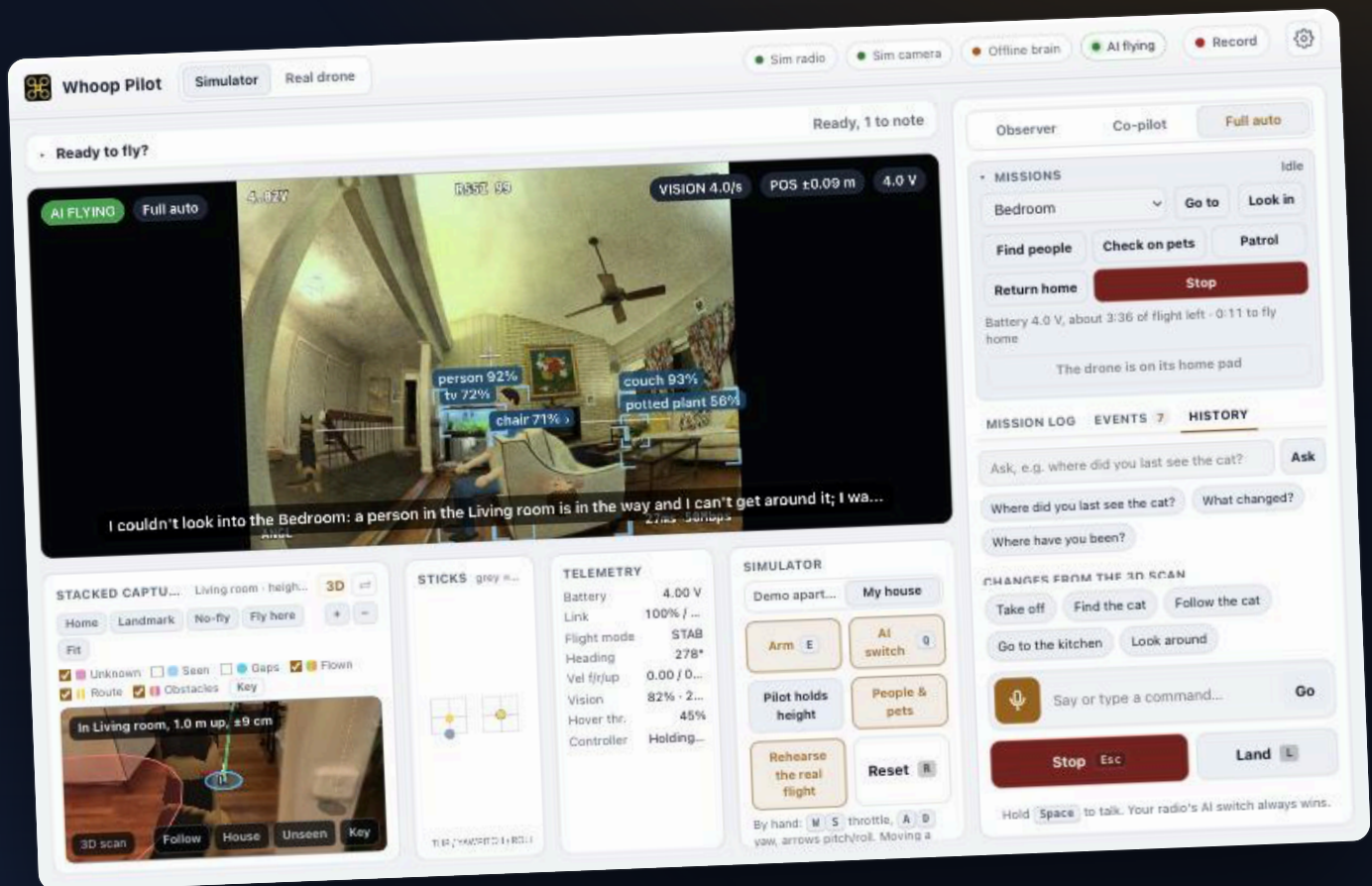


THE FIELD GUIDE



Whoop Pilot

Experimental voice-assisted FPV research.
Supervised use only; start in the simulator.



WELCOME

What Whoop Pilot is

Whoop Pilot is experimental software for supervised FPV research on a Mac. Say “find the cat”, “look in the bedroom” or “go home”, and it plans a route through your house, steers around walls and furniture, and tells you what it saw.

IT CAN

- Understand plain speech (with Claude) or simple typed commands.
- Fly **missions** in your house: go to a room, look in a room, find people, check on pets, patrol, return home.
- Know where it is by matching its live video against the 3D scan of your house.
- Notice things that changed since the scan (a new box, a closed door) and fly around them.
- Remember every flight: where it went, who and what it saw.
- Let you rehearse every mission first in a simulator of *your own* house.

IT IS NOT

- **Not a security camera.** Each flight is a supervised sortie of about 3 minutes (the 480 mAh pack gives about 4).
- **It never arms the drone.** A person puts in the battery and arms with the radio.
- Not hands-off: the Mac, radio and goggles stay on, and the app window must stay visible while it flies.
- Not for places it hasn’t scanned: it only flies where your 3D scan has seen.

Inside this guide

01	How it fits together & safety	3	08	The map and the 3D view	11
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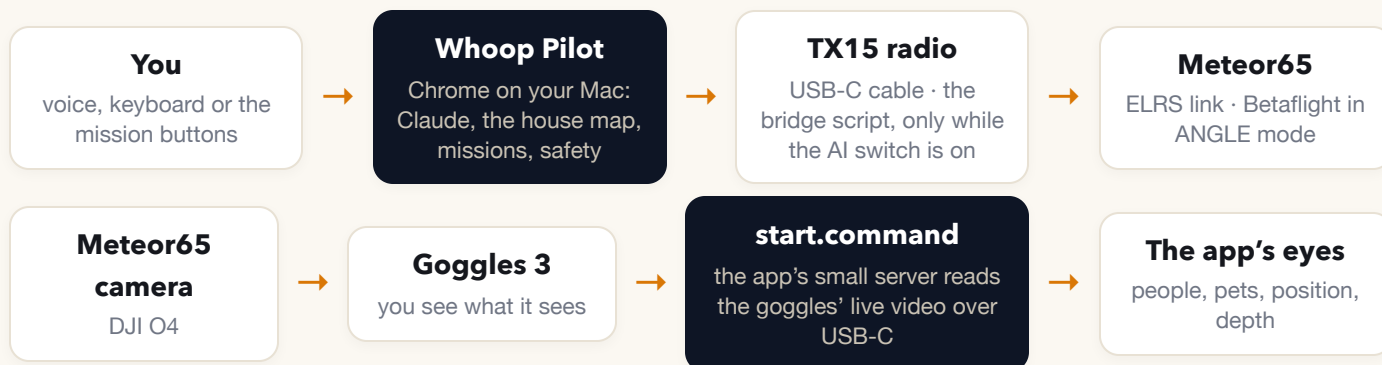
THE SHORT VERSION

Import your house → build the 3D map → set the home pad → rehearse in the simulator → record one flight by hand → calibrate the camera and check it → measure braking → your first mission in **Co-pilot**. Page 14 walks you through it.

01 · THE BIG PICTURE

How it fits together

Your Mac does the thinking. The radio does the flying. Your hands stay in charge.



Three ways to take over, any time

Any stick

Move a stick and that axis is yours again at once.

AI switch

SD on the TX15. Up = you fly.
Down = the AI may fly.

Stop

The red **Stop** button or **Esc**.
Land or **L** to land.

SAFETY RULES

- **The app never arms the drone**, and the radio script can't. You arm with the radio's own arm switch.
- If the app goes quiet for 0.3 s, the radio script **fails safe**: sticks centre and the drone sinks slowly.
- Keep the app window **visible**. Browsers slow hidden tabs, so a mission stops if you hide it.
- The DJI O4 overheats after about **2.5 minutes on the ground** with the battery in. Don't dawdle on the pad.
- For the first flights, keep **people and pets out** of the rooms you fly in, and props away from faces.
- The AI flies in **ANGLE (self-level) mode** only. The radio forces it while the AI switch is on.

02 · DO THIS ONCE

One-time setup

About 20 minutes. Nothing to solder, nothing to buy. The full step-by-step is in the app: the **Setup guide** [↗](#) link.

1 • The radio (TX15)

- 1 Plug the TX15 into the Mac, choose **USB Storage**, and copy **aibrg.lua** into **SCRIPTS / MIXES** on its SD card.
- 2 **SYS** → **Hardware** → **Serial ports** → **USB-VCP**: set it to **LUA**.
- 3 In your Meteor65 model: **Model Settings** → **Enabled Features**, turn **Mixer Scripts** on.
- 4 Add **aibrg** on the Mixer Scripts page. Inputs: the model's Ail/Ele/Thr/Rud, AI switch **SD**, push-to-talk **SF** (optional).
- 5 Add the four **REPL** mix lines for CH1–CH4 and the **ANGLE** line, as shown in the setup guide.
- 6 **Bench check, props off**: SD up = everything as before. SD down with the app closed = sticks still work, flight mode jumps to **ANGLE**.
- 7 Plug USB-C into the Mac and choose **USB Serial (VCP)**.

2 • The video (Goggles 3)

- 1 Install Node.js once: `brew install node`.
- 2 Double-click **start.command**. Keep its window open: closing it stops Whoop Pilot. Chrome opens the app.
- 3 Goggles menu: **Settings** → **About** → **OTG Wired Connection** off.
- 4 **Share Liveview to Mobile Device via Wi-Fi** on. Despite its name, it also turns on the USB live view.
- 5 Connect the goggles' USB-C port to the Mac with a **data** cable (charge-only cables won't work).
- 6 Set the O4 camera to **4:3**.
- 7 In the app, choose **Real drone**. The video appears by itself.

3 • Betaflight / ELRS

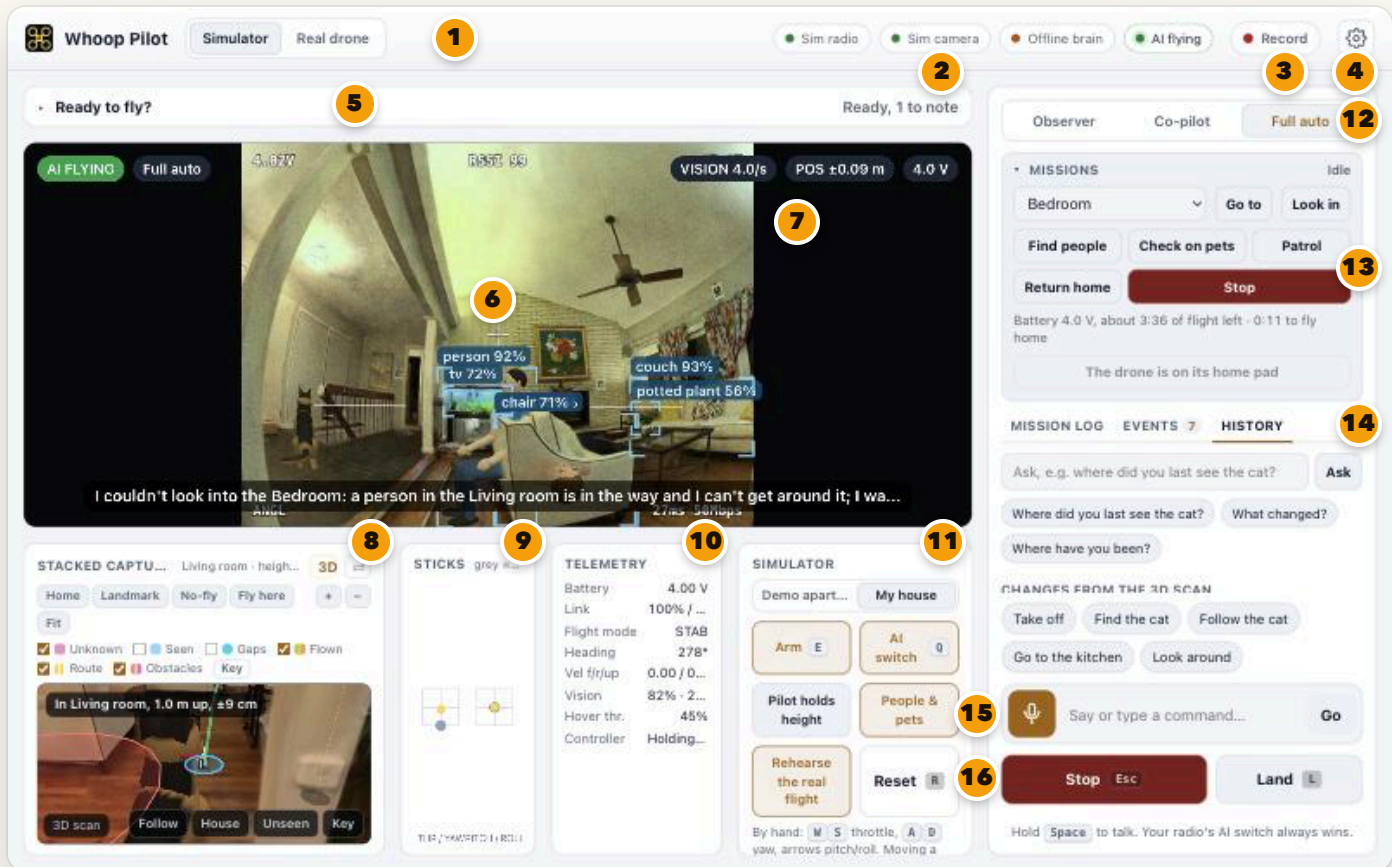
- A switch for **ANGLE** mode (or use the bridge's **ANGLE** line).
- ELRS telemetry ratio **1:8** or 1:4, then **Discover new sensors** once on the radio's Telemetry page.

OPTIONAL · CLAUDE

Add an API key in **Settings** → **Brain** for real language understanding and picture checks. Without a key, a simple offline parser handles the basic commands and every mission still works. Your key stays in this browser and calls go straight to Anthropic.

03 · FIND YOUR WAY AROUND

A tour of the screen



- 1 Simulator / Real drone.** Practise in the simulator; switch to the real drone when ready.
- 2 Status lights.** Radio, goggles, brain (Claude or offline) and the AI switch. Green is good.
- 3 Record.** Saves a flight's video, telemetry and commands on this Mac.
- 4 Settings.** Brain, House, Flight, Camera, Voice, Alerts, Memory, Recordings.
- 5 Ready to fly?** What's still missing before a mission. Click to open.
- 6 Live camera** with what it recognises: people, pets, furniture.
- 7 Flight readout.** Vision fixes per second, position accuracy (POS ±), battery.
- 8 Map** of your house, with tools, layers and a 3D button.
- 9 Sticks.** Grey = you, yellow = what goes to the drone.
- 10 Telemetry.** Battery, link, flight mode, heading, speed.
- 11 Simulator controls.** World, arm, AI switch, people & pets, rehearsal, reset.
- 12 Flying mode.** Observer, Co-pilot or Full auto (page 6).
- 13 Missions.** Go to, Look in, Find people, Check on pets, Patrol, Return home, Stop.
- 14 Mission log · Events · History.** What it's doing, what it found, what it remembers.
- 15 Command bar.** Hold the mic (or **Space**) to talk, or type.
- 16 Stop (Esc) and Land (L).** Always one click away.

04 · WHO FLIES WHAT

Flying modes

Observer

The AI watches and talks, and never moves the drone. Good for checking what it sees.

Co-pilot (default)

You hold the height with the throttle and take off and land when it asks. The AI steers. Recommended starting mode; not a guarantee of safe flight.

Full auto

The AI does everything, including take-off and landing. A whoop has no height sensor, so height comes from the camera. Only after two good Co-pilot flights.

Talking to it

Hold **Space** (or the mic button, or the radio's **SF** button) and speak. Or type in the command bar and press **Go**. The chips above the bar are one-click examples.

WITH YOUR HOUSE LOADED

“Go to the living room”

“Look in the bedroom”

“Find people” · “Check on the pets”

“Patrol the house”

“Come home” · “Land”

“Where are you?”

“Where did you last see the cat?” · “What changed?”

ANYWHERE

“Take off”

“Turn left 90” · “Move forward a metre”

“Look around and tell me what you see”

“Find the cat” · “Follow the cat”

“Stop” · “Land”

GOOD TO KNOW

- With a house map, motion is planned, not improvised: Claude picks the mission, the planner flies it.
- Missions check the battery before every leg and come home with a margin.
- Without a Claude key, the offline brain understands simple phrases like these. With a key, you can talk normally.
- After each command, the log shows what Claude cost (usually a fraction of a cent, thanks to caching).

05 · TEACH IT YOUR HOME

Your house and its 3D map

Whoop Pilot uses the 3D capture of your house (from SiteSpec or Spatial) as its map and as a 1:1 simulator. It only ever flies through space the scan has actually seen.

- 1 **Settings** → **House** lists your captures, newest first. Click **Use this capture**. It copies the capture into the browser, finds the rooms and doors, and builds the flight map.
- 2 Click **Build the 3D map** (about 10 seconds). Your house becomes 5 cm cubes: **clear**, **blocked** or **never seen**. Never-seen space counts as a wall.
- 3 Read the **coverage** table. It says how much of each room the drone can fly in, and which spots the scan missed.

3D map and coverage

Build the 3D map again

Built: 94% of the space the drone can fly in is known. The drone never flies where the scan didn't see.

94%

of the space the drone can fly in (0.3 to 1.6 m up) is known

Good: nearly all of it is known.

The drone only flies through space the 3D scan (or an earlier flight's camera) has seen. Space nobody has seen counts as a wall, so gaps make rooms smaller or unreachable. Rescan a gap with the scanner app, then import the capture again.

Room	Known	Unknown	What to do
Living room	92% known	8% unknown (2.7 m³)	Well covered: 28 gaps worth a rescan (optional)
Room 2	98% known	2% unknown (0.1 m³)	Well covered: 5 gaps worth a rescan (optional)
Room 3	99% known	1% unknown (70 litres)	Well covered: 1 gap worth a rescan (optional)

known open space, unknown (counts as a wall): shares of each room's open space 0.3 to 1.6 m up, furniture left out.

WHAT TO RESCAN (OPTIONAL)

Walk the scanner slowly around the sofa in Living room low, at about 0.7 m (the scanner rode higher than the drone flies), pointing it behind and under it. (7 gaps, 4 spots)

Show on map

Walk the scanner slowly around the table in Living room at knee height (about 0.5 m), pointing it behind and under it. (6 gaps, 2 spots)

Show on map

Walk the scanner slowly around the storage in Living room low, at about 0.7 m (the scanner rode higher than the drone flies), pointing it behind and under it. (5 gaps, 2 spots)

Show on map

Walk the scanner slowly around the chair in Living room low, at about 0.7 m (the scanner rode higher than the drone flies), pointing it behind and under it. (3 gaps)

Show on map

3 more

Every gap (34)

Coverage: 94% of the space the drone can fly in (0.3 to 1.6 m up) is known. Each room gets a plain-language rescan suggestion.

RESCANNING IS OPTIONAL

Gaps just make rooms a little smaller: the drone flies around them. To fill them, walk the scanner slowly around the furniture it names, held low (about 0.7 m: the drone flies lower than you carried the scanner), then import the capture again. Your edits (home pad, names, no-fly zones) are kept.

From above. **Green**: known clear. **Pink**: never seen. Circles are the biggest gaps; squares show where to stand to rescan.

05 · Your house and its 3D map

7

06 · MARK IT UP

Home pad, places and no-fly zones

The home pad

Every flight starts and ends here, and it's how the drone learns where it is. Use the map's **Home** tool: click the spot, drag to set the way the nose faces. Print the pad mat from the setup guide and tape it there, arrow matching the map.

Clearance matters. The app wants 0.5 m clear all round and refuses take-off under 0.35 m. Take-offs can drift 30–45 cm, so keep bags and boxes away from the pad.

Places and no-fly zones

- **Landmark:** click a spot and name it ("cat bed"), then say "go to the cat bed".
- **No-fly:** click the corners of a zone the drone must never enter.
- **Fly here:** click anywhere to send it there.
- Rename rooms in **Settings** → **House**. Ceiling fans and stairs are no-fly automatically.

Let Claude survey the house optional

Claude looks at pictures of your 3D scan, room by room, and suggests room names, places you can send the drone to ("the sofa") and hazards for a small drone (hanging lamps, plants, cables, glass) as no-fly zones.

Nothing changes until you accept it.

WHAT FLIGHTS ADDED

No flight has added to the 3D map yet. With live depth on and the camera calibrated, each flight fills in space the scan missed and marks what its camera looked at.

Check again

Claude's survey of the house

Claude looks at pictures of your 3D scan, room by room, and suggests room names, landmarks you can send the drone to ("the sofa"), and hazards for a small drone (ceiling fans, plants, cables, glass) as no-fly zones. Nothing changes until you accept it.

About \$0.20 for 3 rooms (26 pictures); at most \$0.24 (38 pictures) if the floor or ceiling needs more views.

► Per room

Starting sends these pictures of your house (renders of the 3D scan, no people) to Anthropic's Claude API with your API key, using claude-opus-5. It takes a few minutes; you can stop it.

Ask Claude to survey (about \$0.20)

Camera calibration and the position database

Build the position database It is built from the 3D scan once vision position is on: with the real drone, or in the simulator with Position from its own camera.

Prepare for the flight (before the battery)

The vision models (about 250 MB, once), the 3D map and the position database, in one go: a minute or two the first time, seconds after.

Not calibrated: using the O4's standard lens (127° × 95°, tilted up 20°). Vision position fixes get wider error bars until it is.

The drone finds its position by matching its camera picture to the 3D scan, so the app must know the camera's lens exactly. Calibrate once, and again if the camera's tilt changes.

FROM A RECORDED FLIGHT **RECOMMENDED** **LIVE ON THE PAD**

Record a flight (Record, at the top) that starts

1. Battery in, goggles on, the drone on its home pad: press "The drone is on its

1 · The cost comes first: here about \$0.20 for 3 rooms.

LANDMARKS places you can send the drone to

☒ **sofa**
Room 2 · 86% sure
More than one "sofa" here: name each (say "sofa by the window") to send the drone to it.

☐ **sofa**
Room 2 · 86% sure
More than one "sofa" here: name each (say "sofa by the window") to send the drone to it.

☒ **desk**
Living room · 86% sure

☐ **lamp**
Living room · 55% sure
More than one "lamp" here: name each (say "lamp by the window") to send the drone to it.

☐ **lamp**
Living room · 55% sure
More than one "lamp" here: name each (say "lamp by the window") to send the drone to it.

HAZARDS proposed no-fly zones

☒ **Hanging lamp**
Living room · 78% sure
A pendant lamp hanging low. No-fly zone 1.8 m across, 2.2 to 3.4 m up.

Put 7 on the map **Discard**

2 · Tick what you agree with, rename duplicates, then **Put ... on the map**.

07 · PRACTISE, THEN FLY

Simulator, missions and the HUD

Rehearse in your own house

- 1 **Simulator** → **My house**. The drone camera is rendered from your scan.
- 2 Turn on **Rehearse the real flight**: it flies exactly as the real drone will, with its camera's blur, real speed limits and position from video.
- 3 Press **Reset** **R**, pick a room and press **Look in**, then **Return home**.
- 4 Turn on **People & pets** to see how it waits for people. It keeps 1.5 m away from anyone walking.

By hand in the simulator: **E** arm · **Q** AI switch · **W S** throttle · **A D** yaw · arrow keys pitch and roll.

The mission buttons

Go to	Flies to the room or place picked in the list.
Look in	Stops at the doorway, looks around, reports who and what is there.
Find people	Searches room by room until everyone is found.
Check on pets	Visits the rooms to find the cat and the dog.
Patrol	Every room, then home. Alerts you about people and changes.
Return home	Back to the pad and lands.

Reading the flight readout

AI FLYING **Full auto**

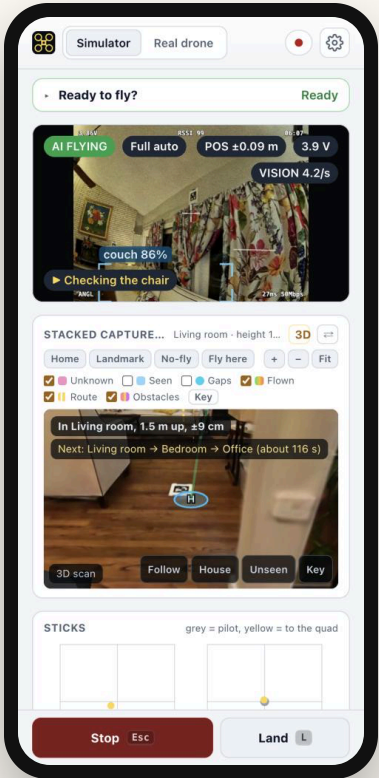
Who is flying, and in which mode.

VISION 4.0/s

Position fixes from the video each second. 3 or more is healthy.

POS ±0.09 m

How sure it is of its position. Small is good.



Works on a phone-sized window too.
Stop and Land stay at the bottom.

07 · IN THE AIR

What it tells you while flying

The drone always says why it's slow, holding or landing: on the video and in the mission log.

YOU SEE	WHAT IT MEANS, AND WHAT TO DO
NO POSITION · SET ON PAD	It doesn't know where it is. Land, put it on the pad, press The drone is on its home pad .
POSITION LOST · HOLDING	It holds still and asks for help (Co-pilot) or lands where it is (Full auto).
NO VISION FIX · VISION DISAGREES	The camera can't place itself right now. It slows down and turns toward a view it recognises.
flying slowly: no live depth yet	It's being careful until its depth estimate starts. Not stuck.
Holding back: a person is close	It keeps 1.5 m from people. It waits, then finds another way or gives up politely.
Waiting for live depth to start before I fly on	A few seconds after a Full-auto take-off. If depth never starts, it lands on the pad.
The way home is blocked (...)	It takes a closer look, tries again with more room, and otherwise lands on the nearest clear spot and says where.
The battery is too low for that	It flies home and lands.
The video is too slow for me to fly by. Please take over and land.	Take the sticks and land.
1 possible change · check	Something the scan doesn't have. Tap it to see the two pictures and decide (page 12). Until then it flies around it.

SPEED LIMITS

The real drone flies at most **0.3 m/s** until you measure its braking, **0.4 m/s** after, and **0.22 m/s** whenever its live depth can't see. Slow is deliberate: it is a 36 g drone indoors.

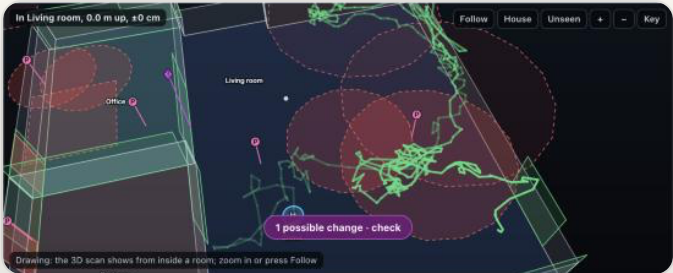
08 · WHERE IT IS, HAS BEEN AND WILL BE

The map and the 3D view

Turn layers on and off under the map, and press **Key** for the legend. Press **3D** to fly the view around your scan.



The map after a patrol: the green trail is where it flew, red circles are where people were, the purple pin is a possible change.



The same flight in 3D. **Follow** keeps the drone in view, **House** shows it all, **Unseen** shows the gaps in pink.

Where it is

The drone icon, with a ring showing how sure it is: yellow when sure, orange when unsure, red when lost.

Where it has been

The trail, kept for every flight, with X where it lost its position. Replay any old flight from History.

Where it will be

The route line through the doors to its goal, redrawn whenever it has to go around something.

Map layers

Unknown	Never seen by the scan: never flown through.
Seen	What the drone's camera has looked at.
Gaps	Spots worth a rescan.

Flown	The trail: where it has been.
Route	The planned path: where it will go.
Obstacles	People, new things, possible changes.

MAP TOOLS

Home sets the pad · **Landmark** names a place · **No-fly** draws a zone · **Fly here** sends the drone · **+ - Fit** zoom · **↔** swaps the map and the camera view.

History, memory and changes

LOGEVENTSHISTORY

Ask, e.g. where did you last see the cat?

Ask

Where did you last see the cat?

What changed?

Where have you been?

I last saw the cat in Room 2 near the step up to entry, 25 min ago (while asked to look in Room 2).

Show on map

CHANGES FROM THE 3D SCAN 1

A new obstacle in the Living room near the chair, about 45 cm across

Not decided yet · last seen 26 min ago

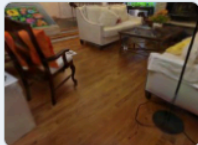
A new obstacle in the Living room near the chair, about 45 cm across

Not decided yet

×

Seen 3 times, last 26 min ago. Living room.

What the drone saw

What the 3D scan shows

Claude: a real change (cardboard box), 86% sure. A brown box on the floor that the 3D scan doesn't have. It suggests confirming it.

Until someone decides, the drone flies around it as if it were there.

Confirm: it's there

Dismiss: not real

It was a person or pet

Ask Claude again

Show on map

Ask Claude sends these 2 pictures to Anthropic's Claude API with your API key (about \$0.02); Claude only advises.

▶ Decided (3)

LAST SEEN

A person

A person in the Living room near the chair, yesterday at 18:10 (on patrol; Claude confirmed).

Cat

The cat in Room 2 near the step up to entry, 25 min ago (while asked to look in Room 2).

Dog

The dog probably in Room 3 near the sofa, yesterday at 18:10 (on patrol).

FLIGHTS 3

Today

The **History** tab is the drone's memory of your house.

Ask it

Type a question or tap a suggestion: “Where did you last see the cat?”, “What changed?”, “Where have you been?”. It answers from what it actually saw, and Claude can use the same memory.

Last seen

People, the cat, the dog and named things, with a picture, the room and when.

Flights

Every flight: time, distance, rooms, findings. Click one to replay its trail on the map and in 3D.

Changes from the 3D scan

Each possible change shows **what the drone saw** next to **what the scan shows**, and what Claude thinks if you allowed it to look. Then you decide:

Confirm: it's there	Added to the map for good. The drone avoids it.
Dismiss: not real	Forgotten. The drone flies there again.
A person or pet	Not a change. It was just someone passing.
Ask Claude again	A second opinion, about \$0.02.

You can do this mid-flight too: tap the change on the map or the button on the video.

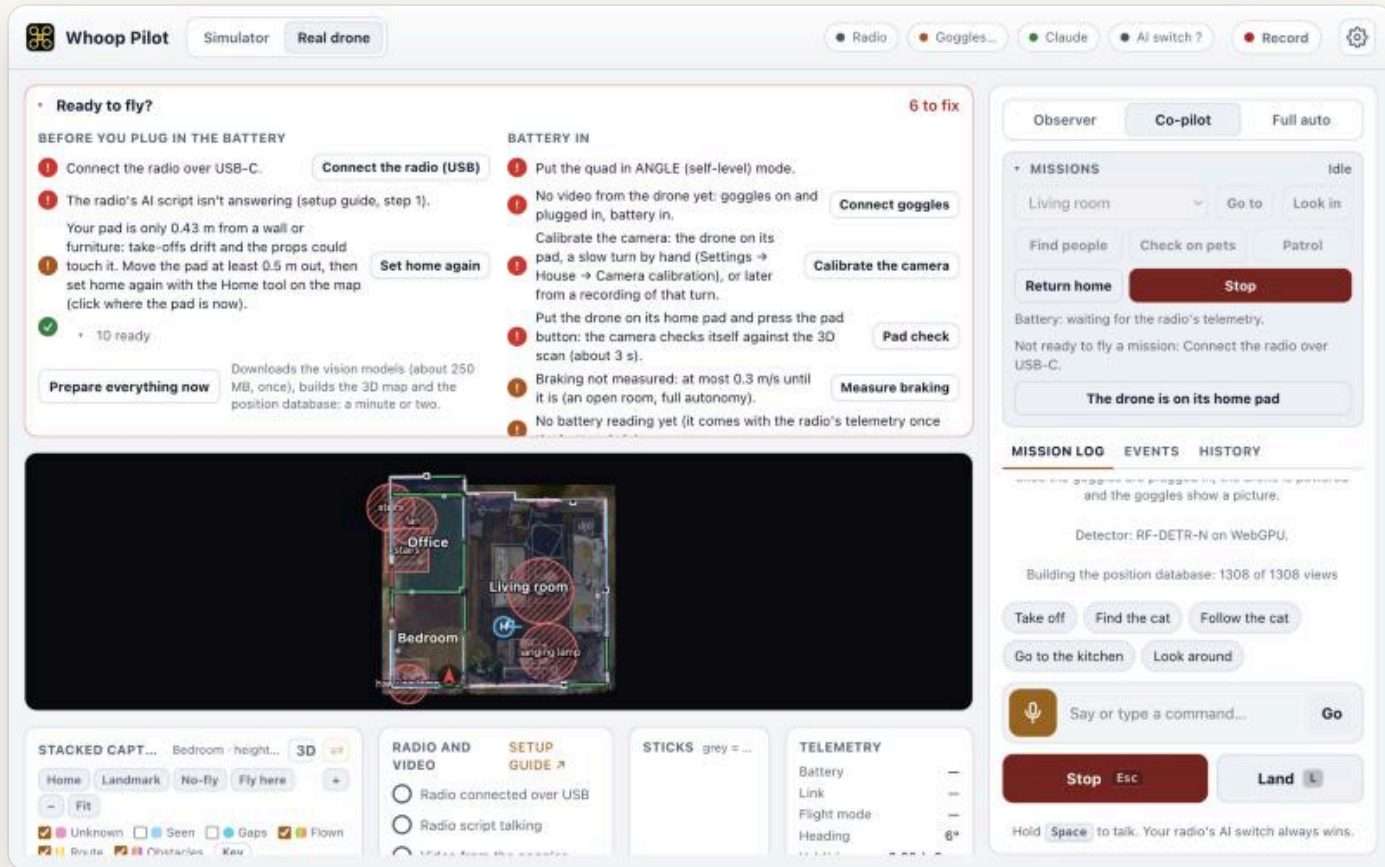
PRIVACY

History is stored locally. AI commands, approved checks and surveys can send images to Anthropic. **Settings** → **Memory** sets how long, and can delete pictures or whole flights.

10 · BEFORE EVERY FLIGHT

Ready to fly?

In **Real drone** mode this checklist opens first. Red items stop missions; orange ones are warnings. Most have a button that fixes them.



Split into what to do **before** the battery goes in, and what needs it in.

Before you plug in the battery

- Radio connected over USB-C, its script answering.
- Home pad set, with room around it.
- **Prepare everything now**: the vision models (about 250 MB, once), the 3D map and the position database. A minute or two.

With the battery in

- ANGLE mode, video from the goggles.
- Camera calibrated, then the **pad check**.
- Braking measured (optional: until then it stays at 0.3 m/s).
- A clock shows the O4's time left on the ground.

11 · THREE BATTERIES

Your first real flight

Three short packs, in this order. With every new battery: drone on the mat, then press **The drone is on its home pad**.

A THE DAY BEFORE · AT THE MAC, NO BATTERY

Get everything ready

Import the house and build the 3D map. Set the home pad and tape the mat down. Rehearse **Look in** and **Return home** in the simulator. Switch to **Real drone** and press **Prepare everything now**. Tidy the floors, lights on, people and pets out. Charge 3 batteries and the goggles.

B PACK 1 · ABOUT 2 MINUTES · YOU FLY

Record a pad turn and a short hover

- 1 Press **Record before** the battery goes in.
- 2 Drone on the mat, nose at the arrow. Battery in, goggles on. Press the pad button.
- 3 Turn the drone slowly by hand on the mat, one full circle (about 20 s), then hold still 3 s. This turn calibrates the camera.
- 4 AI switch **off**. Arm, and fly by hand: waist height, a gentle minute, looking at furnished walls.
- 5 Land on the mat, disarm, press **Record** again. Battery out.

The O4 overheats after about 2.5 minutes on the ground. Don't pause between steps.

C AT THE MAC

Calibrate the camera and check the recording

Settings → **House** → **Camera calibration** → **From a recorded flight** → **Calibrate from it** → **Save it**.

Then **Settings** → **Recordings** → **Check localization**. It passes with at least 3 fixes a second, never lost, and landing within 10 cm of the pad. **Don't go on until it passes**. If it fails, the report says why in one line.



Camera calibration: the yellow dots show how the real lens bends the picture. **Save it** when it says "Good calibration".

The camera found the drone's position all through this flight.

- ✓ Video: 1905 of 1905 frames decoded (60.6 s of it flying).
- ✓ The flight started on the home pad.
- ✓ Position known from 1 s.
- ✓ Position fixes from the camera: 5 a second (at least 3 needed).
- ✓ The position was never lost.
- ✓ 4 cm typical, 11 cm at worst (95%).
- ✓ The video delay setting (100 ms) fits the turns.

Check localization, an example from a test flight. Illustrative historical example; not a flight-safety certification.

11 · THREE BATTERIES, CONTINUED

Braking, then a real mission

D PACK 2 · THE FIRST TIME THE APP FLIES ITSELF

The pad check, then measure braking

- 1 Drone on the mat, battery in, goggles on. Press the pad button: the **pad check** (about 3 s) matches the camera to the scan.
- 2 Choose **Full auto**, then **Measure braking** (**Settings** → **Flight**). It takes off, flies three short runs at 0.3 m/s in the most open spot, and lands. Thumbs on the sticks.
- 3 If it says “*There isn’t room here to measure my braking*” (it needs a straight 1.6 m), skip it: missions still work at 0.3 m/s.

E PACK 3 · YOUR FIRST MISSION

Look in a room, in Co-pilot

- 1 Choose **Co-pilot**, press **Record**. Drone on the mat, battery in, pad button. No red items in Ready to fly?
- 2 Pick a room next to the pad and press **Look in**. Arm when it asks, flip the **AI switch on**, take off with the throttle to waist height.
- 3 **You hold the height, the app steers.** It asks you to land at the end.
- 4 Afterwards: stop Record, run **Check localization** on it, and look at **History**.

F LATER

When Full auto is OK

- Two Co-pilot missions in a row whose recording check passed, never lost.
- Braking measured, and take-offs that stayed well clear of everything.
- The same mission flown cleanly in **Rehearse the real flight**.

Then try one short **Look in** in Full auto, over carpet, nobody else in the room, thumbs on the sticks. Watch the height most: it comes from the camera.

AFTER EVERY FLIGHT

- Stop **Record** and take the battery out (the O4 keeps heating while powered).
- Glance at **History**: the path it flew, who it saw, any possible changes to decide.
- Run **Check localization** on recorded flights while you’re still learning.

STOP FLYING IF

- The recording check fails, or the drone says it lost its position more than once.
- Take-offs drift toward furniture: move the pad somewhere more open.
- The video stutters or the Mac gets hot and slow: close other apps first.

12 · FIXES

When something is wrong

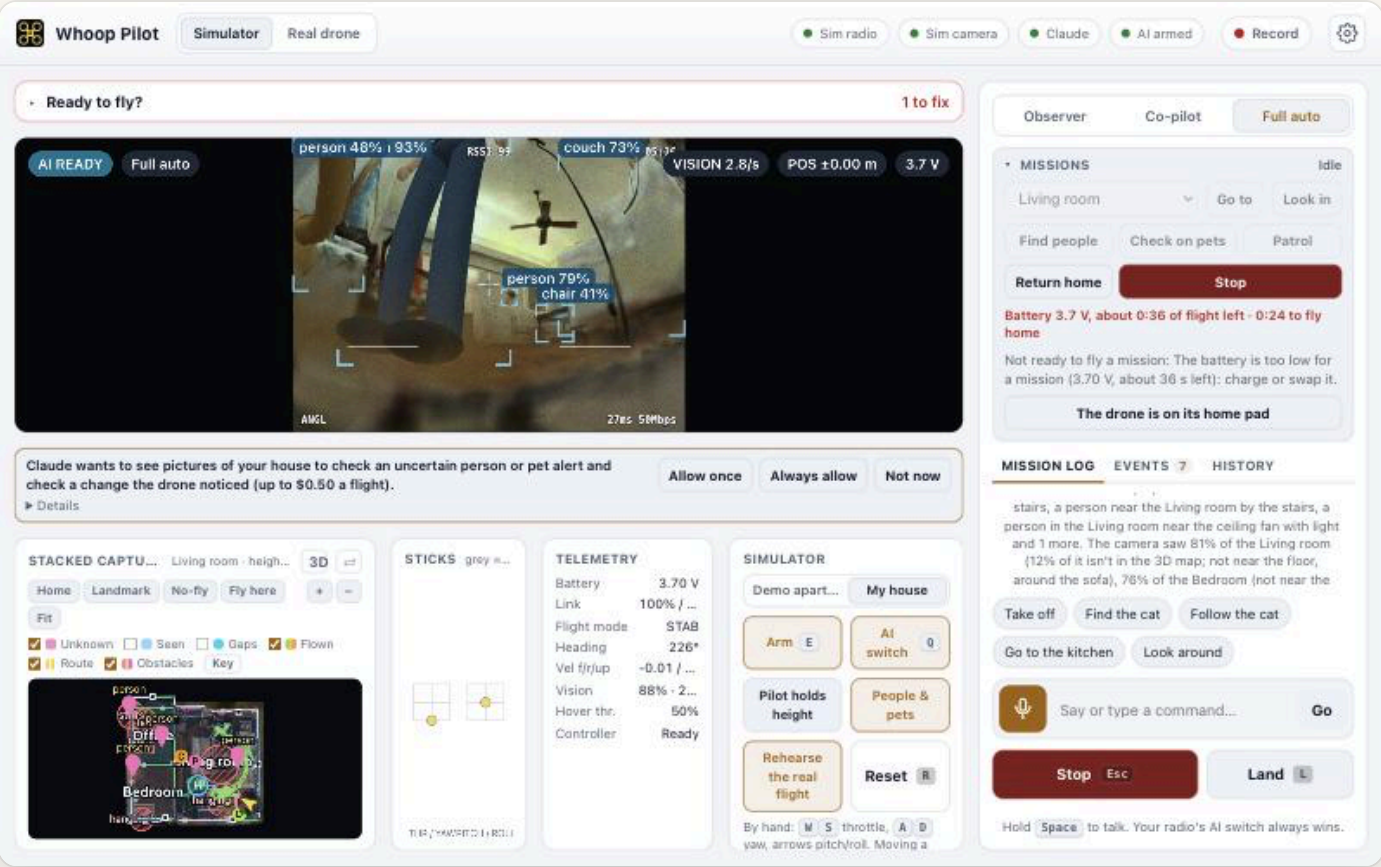
THE APP SAYS	DO THIS
Connect the radio over USB-C · The radio's AI script isn't answering	Plug the radio in, press Connect the radio (USB) . USB-VCP must be LUA and the Mixer Script added (page 4).
Put the quad in ANGLE (self-level) mode	Flip your flight-mode switch to ANGLE .
No video from the drone yet · Still finding the camera picture	Goggles on and plugged in, battery in, wait a few seconds. Set the O4 to 4:3 if it says the shape is wrong.
Your pad is only ... m from a wall or furniture	Move the mat somewhere more open and press Set home again .
Calibrate the camera · Put the drone on its home pad	Steps C and D of the first flight.
Battery in for ...: about ... before the O4 overheats	Hurry, or unplug and let it cool for a few minutes.
The computer is too busy to fly safely	Close other tabs and apps.
No position fix from the camera in the last second	Let the camera see the room, not a blank wall. Or land, put it on the pad and press the pad button.
Braking not measured: at most 0.3 m/s	Fine as it is, or do step D.

Goggles status

Server not running	Double-click start.command . If it complains, install Node.js (brew install node).
No goggles on USB	Use a data cable, goggles on, OTG Wired Connection off .
The goggles don't answer	Drone powered and linked; goggles awake (wear them or cover the sensor); not in Real View or the album; Share Liveview on .
Connected, but no picture	Power the drone and check the goggles show its camera.
Goggles went quiet	Nothing to do: the app reconnects by itself.

Claude, cost and privacy

Claude is the drone’s brain for language and a second pair of eyes. You stay in control of what it sees and what it costs.



AI commands include a camera frame; picture checks and surveys use their consent flow. Review what is sent before using a key.

What gets sent

- **Voice commands:** your words, the drone’s state and the current camera frame.
- **Picture checks,** only with your OK: a possible change (live vs scan) or an uncertain person or pet.
- **The survey:** renders of your 3D scan, with no people in them.

Your choices

Allow once · **Always allow** · **Not now** · **Never**.
Change it any time in **Settings** → **Brain**: picture checks **Ask** (default), **On** or **Off**.

What it costs

A voice command	usually under a cent (prompt caching)
A change check	about \$0.02
The house survey	about \$0.20 for 3 rooms, shown first
Per-flight cap	\$0.50 by default (Settings → Brain)

The log shows the cost of every call, split into voice commands, picture checks and memory questions.

YOUR KEY

Your API key lives only in this browser. Calls go straight to Anthropic. Claude only advises: the planner and safety layer decide how the drone moves.

14 · KEEP THIS ONE HANDY

Quick reference

TAKE OVER

Any stick that axis is yours at once

AI switch SD up you fly everything

Esc / Stop stop the mission

L / Land land now

KEYS

Space (hold) talk

E · **Q** arm · AI switch (simulator)

W **S** · **A** **D** throttle · yaw (simulator)

Arrows · **R** pitch/roll · reset (simulator)

SAY

Go to the ... fly to a room or place

Look in the ... look into a room

Find people search every room

Check on the pets find the cat and dog

Patrol every room, then home

Come home back to the pad

What changed? from memory

EVERY FLIGHT

- 1** Ready to fly? has no red items.
- 2** Record on, battery in, drone on the mat.
- 3** Press **The drone is on its home pad**.
- 4** Pick a mission, arm, AI switch on.
- 5** Thumbs on the sticks. About 3 minutes.
- 6** Land, disarm, battery out. Glance at History.

NUMBERS TO REMEMBER

Flight time about 3 minutes per sortie (4 min pack)

O4 on the ground about 2.5 min before it overheats

Speed 0.3 m/s · 0.4 after braking test

Pad clearance 0.5 m wanted, 0.35 m minimum

Distance from people 1.5 m

Claude budget \$0.50 per flight